: four independent equations.

The satellite clock slows by . The ground reading minus satellite reading is

Here is satellite speed relative to the ground, determined by

is the gravitational constant, Earth's mass, orbit radius. From ,

Here is Earth's radius, altitude and ground gravity. Substitution gives , hence , which is very small. Thus

Finally the clock difference over is

The satellite reading minus the inertial-clock reading at infinity is

The gravitational potential magnitude at the satellite clock is

Thus

Numerically, is very small, so equation becomes

Similarly, ground reading minus the inertial clock at infinity is

The gravitational potential magnitude at the ground clock is